CAPSTONE PROJECT PROPOSAL FORM

Project Title:

SleepEase – Islamic AI Anxiety & Sleep Assistant

Group Number:

47

Supervisor:

Dr. Noor Hasrina Bakar

Industry Client:

N/A (Academic Project Only)

Project Members		
Name	Student ID	Programme
Salman Zaman	0369750	Bachelor of Information Technology
Muhammad Yahya	0359154	Bachelor of Computer Science
Ayham Aejaz	0366634	Bachelor of Computer Science

Liu Jiakang	0369697	Bachelor of Computer Science
KM Tahmid Hassan	0367314	Bachelor of Information Technology

A. EXECUTIVE SUMMARY

SleepEase is a mobile application which aims to assist the user to decrease anxiety and enhance the quality of sleep using faith-based and mindfulness advice with the help of Artificial Intelligence (AI). The application is based on the Islamic values of peace, reflection, and emotional equilibrium, though it is open to both users with other religions or origins. It gives an opportunity to all the people to feel calm emotionally and sleep better by using brief evening rituals and positive thoughts.

Users on the app every night can do a quick check-in in order to document mood or emotions. This input is processed by the AI and presented with the recommendations of Quranic verses or the Hadiths or generic inspirational quotes depending on the mode that the user has chosen. It also offers brief relaxing practices like breathing exercises, gratitude meditations, and relaxing sound.

Two modes are available to the user:

Islamic Mode - Incorporates Quranic and Hadith-based guidance, du'as and reflections.

General Mode - Provides non-Muslim users with neutral reminders of mindfulness, gratitude journaling and meditation routines that relieve stress.

SleepEase is going to be created on Android with Kotlin to implement it easier. Firebase will be used to store the data, and the AI-based mood detection and recommendations will be implemented using Python API. The key aim is to assist users to sleep well, cope with anxiety and form a balanced emotional and spiritual schedule that does not violate various beliefs and preferences. Besides the basic wellness capabilities, the system will incorporate Analytics to

measure user engagement, retention, and behavioral patterns to make continuous improvements. Predictive modelling and compliance monitoring will as well be implemented in order to ensure safe, ethical and data-driven personalization of the user experience.

B. PROJECT PURPOSE

Problem Statement

Emotional overload and lifestyle habits cause most students, professionals, and parents to be stressed, anxious, and unable to sleep. Most of the wellness applications out there are limited to general psychology or medical strategies, excluding the more personal spiritual component most users appreciate. The Qur'an teaches:

> *"Verily, in the remembrance of Allah do hearts find rest."* (Surah Ar-Ra'd, 13:28)

Nevertheless, the number of non-Muslim users who want to find peace and thankful tools in their products also increases. The gap that SleepEase fills is that Muslims have faith-based reflection, whereas other people are offered with the general guidance of mindfulness. This also renders it an ethical, culturally sensitive and inclusive remedy to enhance mental and emotional health.

Project Objectives

1. To develop a mobile application that is artificial intelligence powered and provides spiritual and emotional counsel to users to assist them in reducing anxiety and improving their sleep quality.

2. Implementing custom emotional support features, such as AI-driven chatbots, daily reflections, gratitude journaling or others that help the users to engage regularly and promote emotional well-being.

3. TTo create an Android application which is not only connected to the backend artificial intelligence in a secure manner but also uses Python, Kotlin, and Firebase to control data.

4. To use advanced BI analytics to track how users engage with the app, group them based on their needs, understand their emotions, and use models to predict stress levels and check the accuracy of the content, all to make the app more personal for each user. We achieve this by ensuring ethical reliability, providing long-term effective support, and focusing on commercialization.

C. PROPOSED FUNCTIONALITY

No.	Proposed Functionality	Problems Solved / Opportunities	Objective Fitment
1	**Daily Reflection & Check-In Streak**	Aids users in restoring gratitude and mindfulness by encouraging them to take time and reflect on their day in the form of daily questions and reflection streaks.	Objective no. 01
2	**Interactive Duʿā or Gratitude Wall**	Enables users to save and re-read their most preferred duʿa or personal gratitude notes to be comforted and motivated.	Objective no. 01 & 02
3	**AI Mood-to-Motivation Chat**	Delivers customized spiritual or inspirational messages depending on the emotions of the user to make them feel less stressed and managed.	Objective no. 01 & 03
4	**Progress Dashboard (Emotional &**	Shows mood, reflection streaks and sleep progress, motivating users to be consistent and balanced.	Objective no. 02 & no. 04

	Spiritual Tracker)		
5	**User Retention & Churn Analytics Module**	Examines frequency of user log-in, number of drop-offs and seasonal use (e.g. exams/Ramadan) to determine disengaged users and help enhance long-term app use.	Objective no. 04
6	**User Segmentation & Personalized Insights Layer**	Segment users according to emotional trends and demographic (e.g.: students vs parents vs professionals) to provide reminders, push notifications and AI recommendations driven by user group behaviour.	Objective no. 04
7	**Predictive Well-Being & Compliance Monitoring Engine**	Creates predictive scores like Sleep Improvement Index or Emotion Stability Score and also audits Islamic and motivational content on each score to ensure that it is ethically accurate and safe to users.	Objective no. 03 & no. 04

D. ASSUMPTIONS

1. It is assumed that users will have stable internet access to support AI processing and secure cloud-based data storage throughout the application's operation. This ensures smooth communication between the mobile application, backend server, and database.
2. The source of Islamic content for the application will come solely from verified and authentic platforms such as Quran.com and Sunnah.com in order to ensure the accuracy and reliability of Qur'anic verses and Hadith references.

3. For non-Islamic users for all content will be based on universal values such as gratitude, peace, mindfulness and kindness for inclusivity and cultural sensitivity with users from all sorts of backgrounds.
4. Sentiment analysis will be implemented to identify the emotional state of the user and predefined responses will be generated by AI and customised for each mode Islamic or General to ensure relevant and meaningful recommendations.
5. The application will maintain ethical standards of presentation by ensuring all religious and inspirational materials are presented respectfully and in a culturally appropriate fashion.
6. With user consent, aggregated and anonymized data on user behavior (numbers of sessions, mood entries, interactions with the AI, etc.) will be collected and used for analytics (retention, user segmentation, etc.) and for making predictions

7. Verifications and checks will be put in place to ensure that all AI-generated responses including Islamic content matter are contextually accurate, non-harmful and respectful before being deployed at scale

8. Anonymized interaction information will be analyzed with the tools of Firebase Analytics and Power BI to determine user engagement, user retention trends, and outcomes of mood improvement. All analytical processes will be run under strict privacy and ethical compliance; initial testing will be performed against synthetic (dummy) data to validate that the BI components work well and are robust before running against users' actual data.

E. PROJECT DESCRIPTION, SCOPE AND MANAGEMENT MILESTONES

Project Description

SleepEase is an emotional and spiritual wellness app, which is dual-mode and based on AI technology. Users enter their feelings at night and the system gives them relaxing recommendations. As a Muslim user, one can get Islamic reflections, duaas, and Quranic advice, and as a non-Muslim user, one can get general quotes of mindfulness and gratitude prompts. The AI of the app is trained on the interactions with users to become more personalized with time. It promotes regular emotional reflection, improved sleep patterns, and serenity by using it regularly.

Scope

1. Platform: Android (coded in Kotlin).
2. Target Users: Audience who want to have emotional peace and better sleep.

Core Features:

- ☐ Sleep pre-sleep check-in with AI-based customized instructions.
- ☐ Streak of reflecting every day to promote consistency.
- ☐ Personal duʿā or gratitude wall for saving peaceful notes.
- ☐ AI chat to be motivated or find spiritual guidance.
- ☐ Progress dashboard in order to track emotional and sleep balance.
- ☐ Cloud storage of data through firebase.
- ☐ User retention and churn statistics to track the long-term engagement trends.
- ☐ Persona mining and user segmentation in order to personalize suggestions and notifications on the basis of the various user groups.
- ☐ Well-being scoring and monitoring of ethical compliance to achieve safe, respectful and reliable content delivery.

Out of Scope:

- ☐ IOS application development in Capstone 1.
- ☐ Clinical diagnosis or medical treatment characteristics.
- ☐ Public sharing or social media.

Development Tools and Technologies

To ensure efficient and high-quality development, the following tools and frameworks will be used:

Area	Technology / Platform	Purpose
Frontend (Mobile App)	**Android Studio (Kotlin)**	To develop and create the Android application with a clean and responsive interface.
Backend (Server)	**Python (Flask or FastAPI)**	To deal with AI logic, process data requests, and integrate the mobile app with backend services.
Database & Authentication	**Firebase (Firestore & Auth)**	To store user information like mood records, reflections, duʿās, and sleep logs safely as well as control user privacy.
AI & NLP Processing	**Hugging Face API / TextBlob (Python)**	To analyze the feelings of the users and provide them with suitable recommendations.
Islamic & Motivational Content Sources	**Quran.com**, **Sunnah.com**, and reliable quote databases	To gather genuine verses of the Quran, Hadith, and overall motivational information.
Cloud Hosting (if required)	**Google Cloud / Render / Railway**	To deploy and implement the Flask or FastAPI server to respond in real-time to AI.

Design & UI Tools	**Canva / Figma**	To develop the layout and design prototypes as well as the color schemes of the app.
Version Control	**GitHub**	To maintain and keep the source code and revisions of the project.
Machine Learning for Prediction & Clustering	**Scikit-learn / Python ML**	To come up with churn prediction models, segmentation clusters, and wellness score indexes.
Business Intelligence & Visualization	**Tableau**	To develop analytical dashboard in retention, segmentation and decision insights.
User Analytics & Tracking	**Firebase Analytics**	To win over user engagement, retention, churn and complete lifecycle behavior.
Ethical & Compliance Monitoring	**Excel QA Logs / Power BI Alerts**	To make sure that Islamic and inspirational AI content is correct, safe, and morally sound.

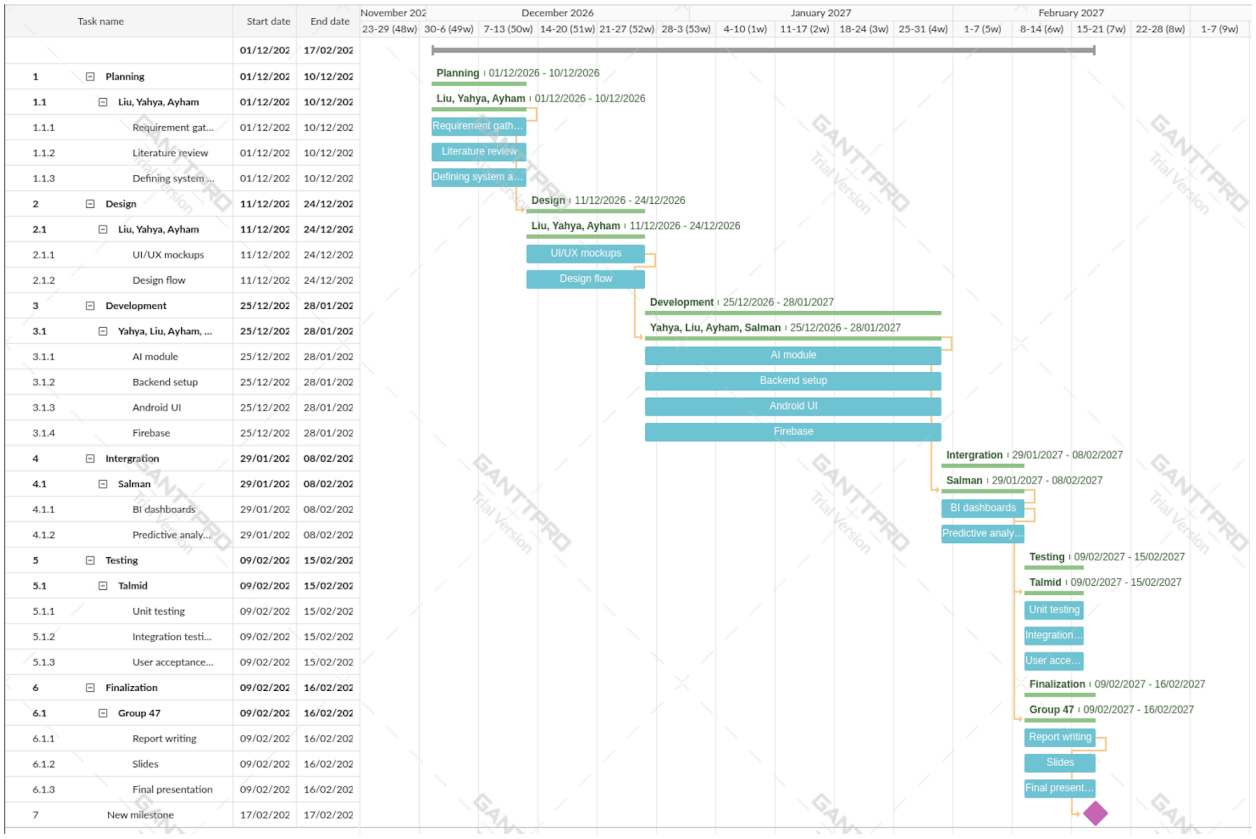
Summary of Milestones and Deliverables

Milestone No.	Milestone	Person Responsible	Duration (days)

001	Requirement gathering and literature review	Group 47	10
002	System architecture and UI mockups	Group 47	14
003	Backend and Firebase database setup	Muhammad Yahya	20
004	Android UI design and integration.	Liu Jiakang	20
005	AI mood detection and recommendation module	Ayham Aejaz	15
006	BI & Predictive Analytics Integration	Salman Zaman	10
007	Testing and debugging	KM Tahmid Hassan	7
008	Documentation and presentation	Group 47	5

Project Development Timeline

The Gantt chart below shows the entire timeline of the SleepEase development process of 70 days. Works are divided into six main phases which are Planning, Design, Development, Integration, Testing and Finalization by making sure that responsibilities and time are clearly defined.

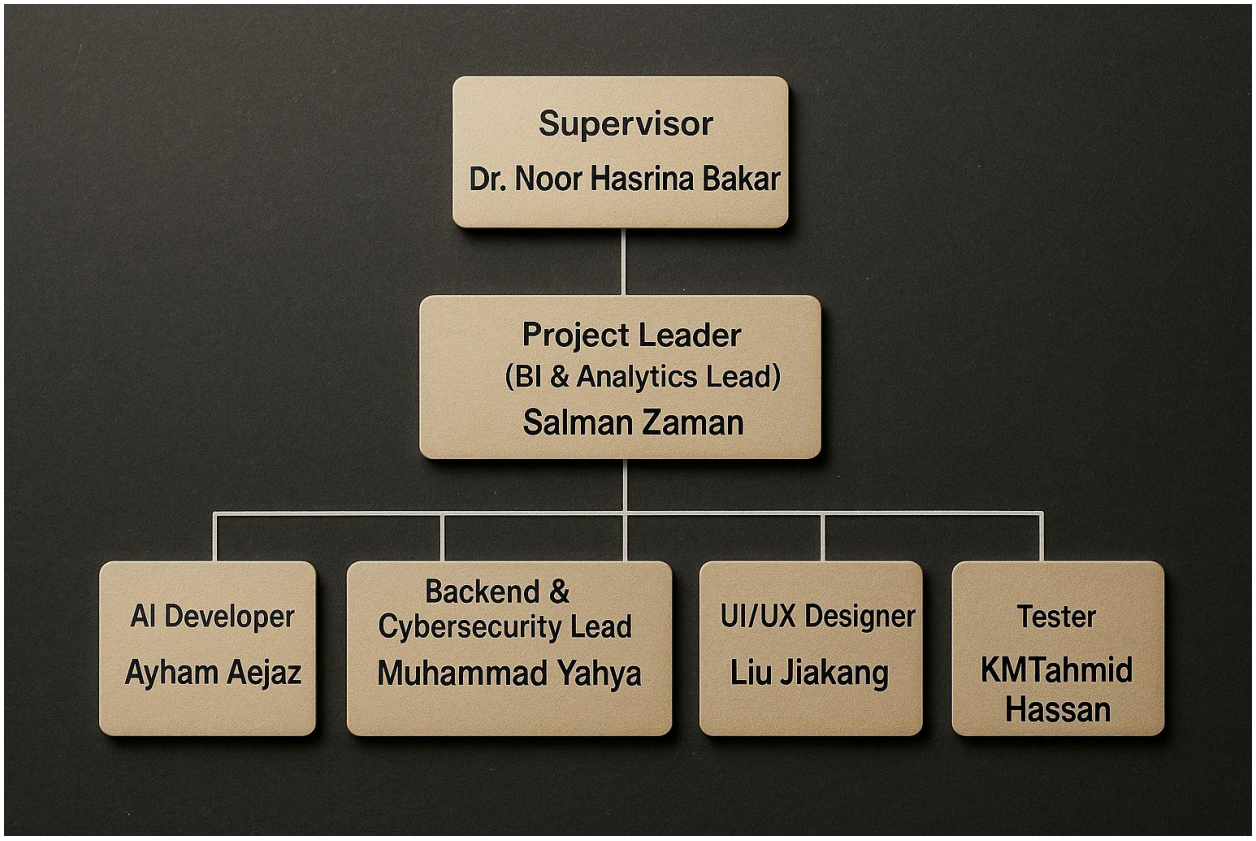


The Planning phase involves gathering the requirements, doing literature review, and defining the system. The Design phase covers UI/UX mockups, and the system flow. At Development, several parallel tasks are implemented like AI module, Backend setup, Android UI and Firebase.

The Integration phase takes place, where BI dashboards and predictive analytics are added. Afterwards there is the Testing phase, which involves unit, integration and user acceptance testing. Finally, the Finalization phase is where the report writing, slides preparation and final presentation by all the group members with the final project completion milestone are accomplished.

F. PROJECT ORGANIZATION

Organization Chart



Organization Description

Group 47 is an academic development team supervised by Dr Noor Hasrina Bakar. Each member has a part in various parts of the project, such as system design, backend development, testing, and analytics. The general goal of the team is to blend the Islamic ethics, inclusivity and modern technology to make a valuable and practical mobile solution that benefits Muslims and non-Muslims.

G. ROLES AND RESPONSIBILITIES

Name	Role	Responsibilities
Dr. Noor Hasrina Bakar	Supervisor	Guides academic and ethical direction, ensuring content accuracy and inclusivity.

Ayham Aejaz	AI Developer	Develops the AI-based mood-detection and recommendation logic using Python and NLP frameworks. Integrates sentiment-analysis models with backend APIs to deliver personalized reflections and motivational responses. Ensures algorithm accuracy, reliability, and ethical content generation for both Islamic and General modes.
Muhammad Yahya	Backend & Cybersecurity Lead	Manages backend architecture and Firebase database to ensure reliable, secure, and efficient data operations. Implements authentication, encryption, and access-control mechanisms to protect user data and maintain confidentiality. Oversees secure API communication between the mobile app, AI services, and cloud infrastructure while ensuring compliance with ethical and privacy standards..

Liu Jiakang	UI/UX Designer	Designs user-friendly interfaces suitable for both Islamic and General modes, focusing on accessibility and visual harmony. Ensures privacy-aware interface layouts that prevent unauthorized display or misuse of user data. Collaborates with backend developer to maintain secure data flow between user interface and Firebase authentication systems.
Salman Zaman	BI & Analytics Lead	Implements user analytics, dashboards, and predictive insights to support data-driven decision-making. Develops data-visualization reports to track retention, engagement, and well-being metrics while ensuring analytics comply with privacy and ethical standards.
KM Tahmid Hassan	Tester	Conducts comprehensive testing to ensure the app operates smoothly, performs efficiently, and maintains accuracy across Android devices. Prepares detailed test reports and verifies that all features meet usability, functionality, and security standards before deployment.

H. APPENDICES

Authentic Quranic and Hadith sources, AI design flow, and user testing results will be attached in the **Capstone Project 2 report**.

G. References

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8. Owens, J., Rassool, G.H., Bernstein, J., Latif, S. and Aboul‐Enein, B.H., 2023. *Interventions using the Qur'an to promote mental health: A systematic scoping review*. *Journal of Mental Health*, 32(4), pp.842–862. https://doi.org/10.1080/09638237.2023.2232449